Project Title:
Brain Tumor Classification Using CNN on MRI Images

Dataset: **Brain Tumor MRI Dataset**

https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

A deep learning-based system that uses Convolutional Neural Networks (CNN) to classify brain MRI images into four categories: Glioma, Meningioma, Pituitary tumor, and No tumor. The model is trained with preprocessing, data augmentation, and regularization techniques. A user-friendly interface allows users to upload MRI images and view predictions with confidence scores.

The model was trained using Google Colab, which provides a powerful GPU environment for faster training. I uploaded the Brain Tumor MRI dataset to Colab, performed preprocessing, and trained a CNN model using TensorFlow/Keras.

After training was completed, the model was saved as a file named brain_tumor_model.keras. This file contains the learned weights and structure of the trained model.

Model Training on Google Colab (train.py)

The same code was uploaded to Google Colab (to use free GPU) and executed there:

1. Data generators load the images in batches
2. The model is trained for up to 20 epochs with 3 important callbacks:
    - EarlyStopping — stops training if validation loss doesn't improve for 20 epochs
    - ModelCheckpoint — saves the best model (by validation accuracy) as brain_tumor_model.keras
    - ReduceLROnPlateau — reduces learning rate by 5× if validation loss plateaus for 5 epochs
3. After training, accuracy and loss plots are saved to the outputs/ folder

The trained model file brain_tumor_model.keras (~231 MB) was then downloaded from Colab and placed in the local outputs/ folder.

Model Architecture (model.py)

A custom CNN (Convolutional Neural Network) is built using Keras Sequential API with 4 convolutional blocks + a fully connected classifier:

Block	Layers	Purpose
Block 1	Conv2D(32) → BatchNorm → MaxPool → Dropout(0.25)	Detect basic edges and textures
Block 2	Conv2D(64) → BatchNorm → MaxPool → Dropout(0.25)	Detect shapes and patterns
Block 3	Conv2D(128) → BatchNorm → MaxPool → Dropout(0.25)	Detect complex features
Block 4	Conv2D(256) → BatchNorm → MaxPool → Dropout(0.25)	High-level feature extraction
Classifier	Flatten → Dense(512) → BatchNorm → Dropout(0.5) → Dense(4, softmax)	Final classification

Streamlit Web App (Frontend/app.py + style.css)

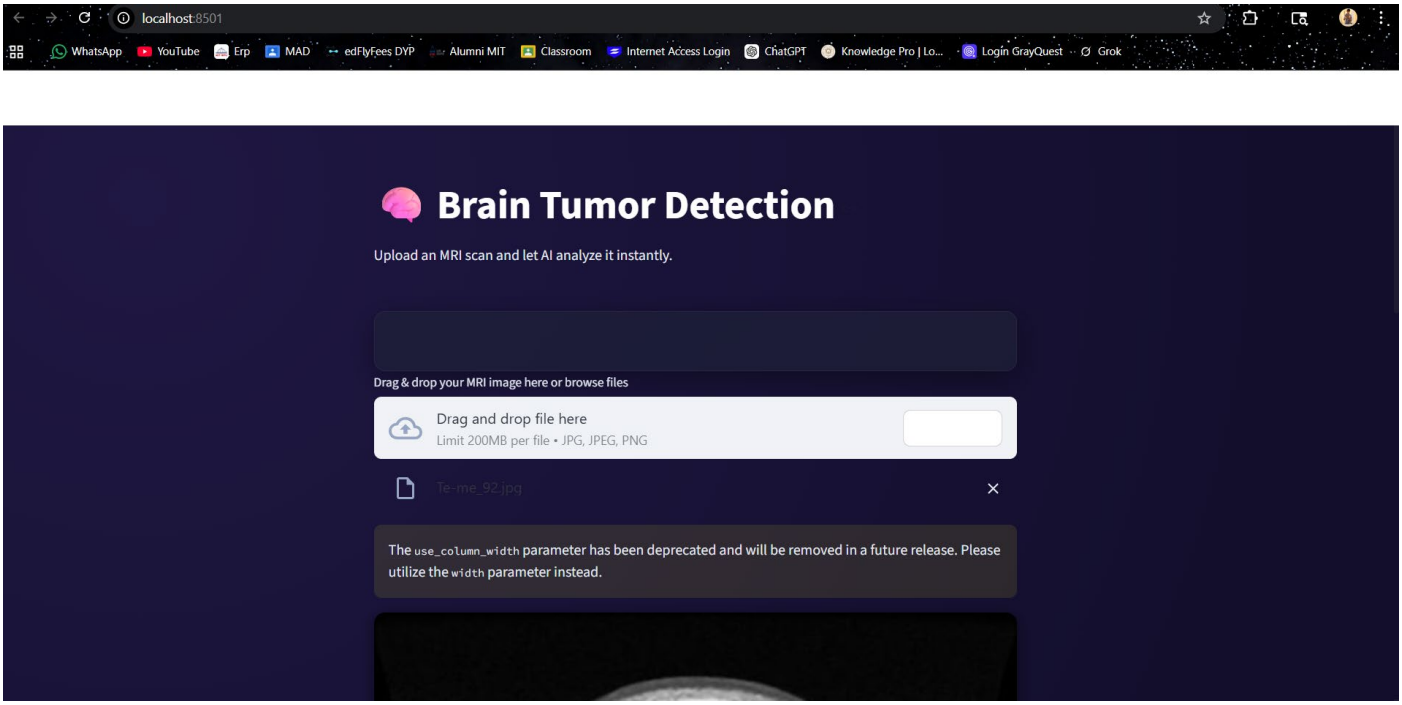
A user-friendly Streamlit web interface is built:

1. Custom dark theme with a purple gradient background, glassmorphism card styling, and gradient buttons (defined in style.css)
2. User uploads an MRI image via a drag-and-drop uploader
3. Clicks "Predict Tumor" button
4. The app loads the trained model, preprocesses the image, and runs the prediction
5. Displays the predicted class (e.g., "GLIOMA") and a bar chart of confidence scores for all 4 classes

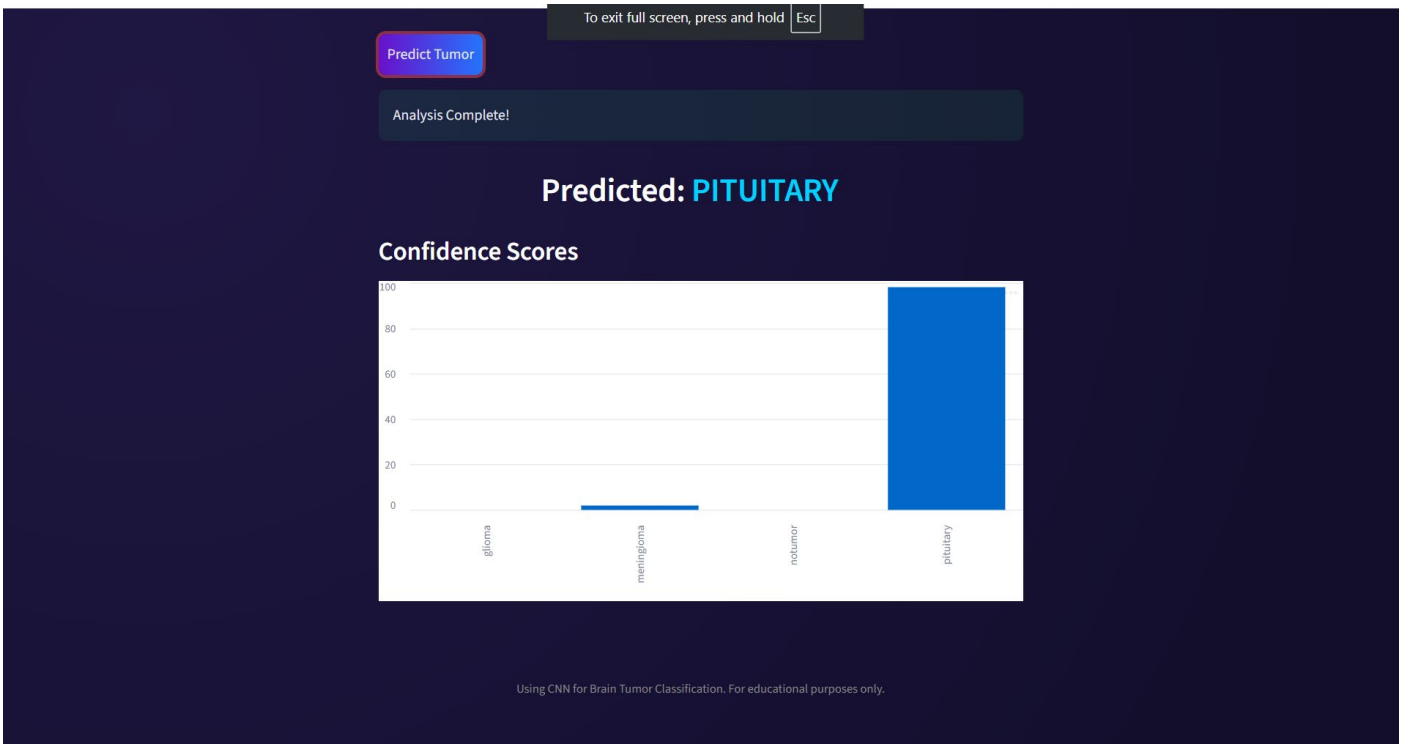
Outputs:

Images are taken from testing folder of dataset

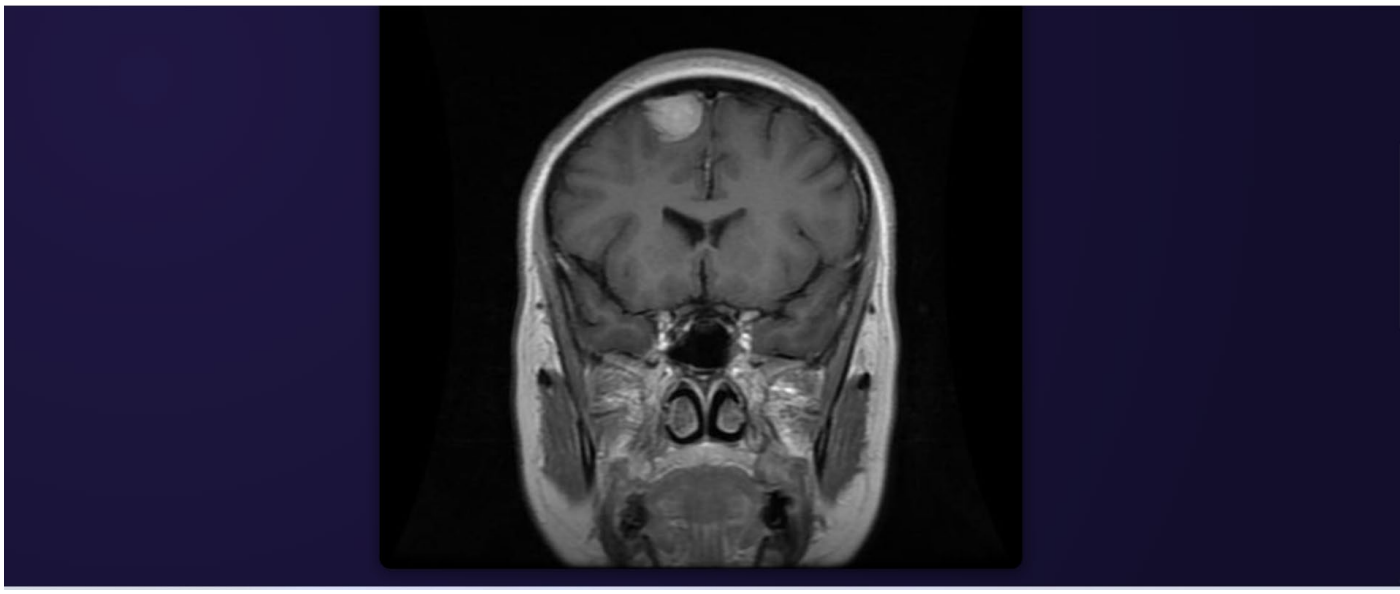
1. Pituitary

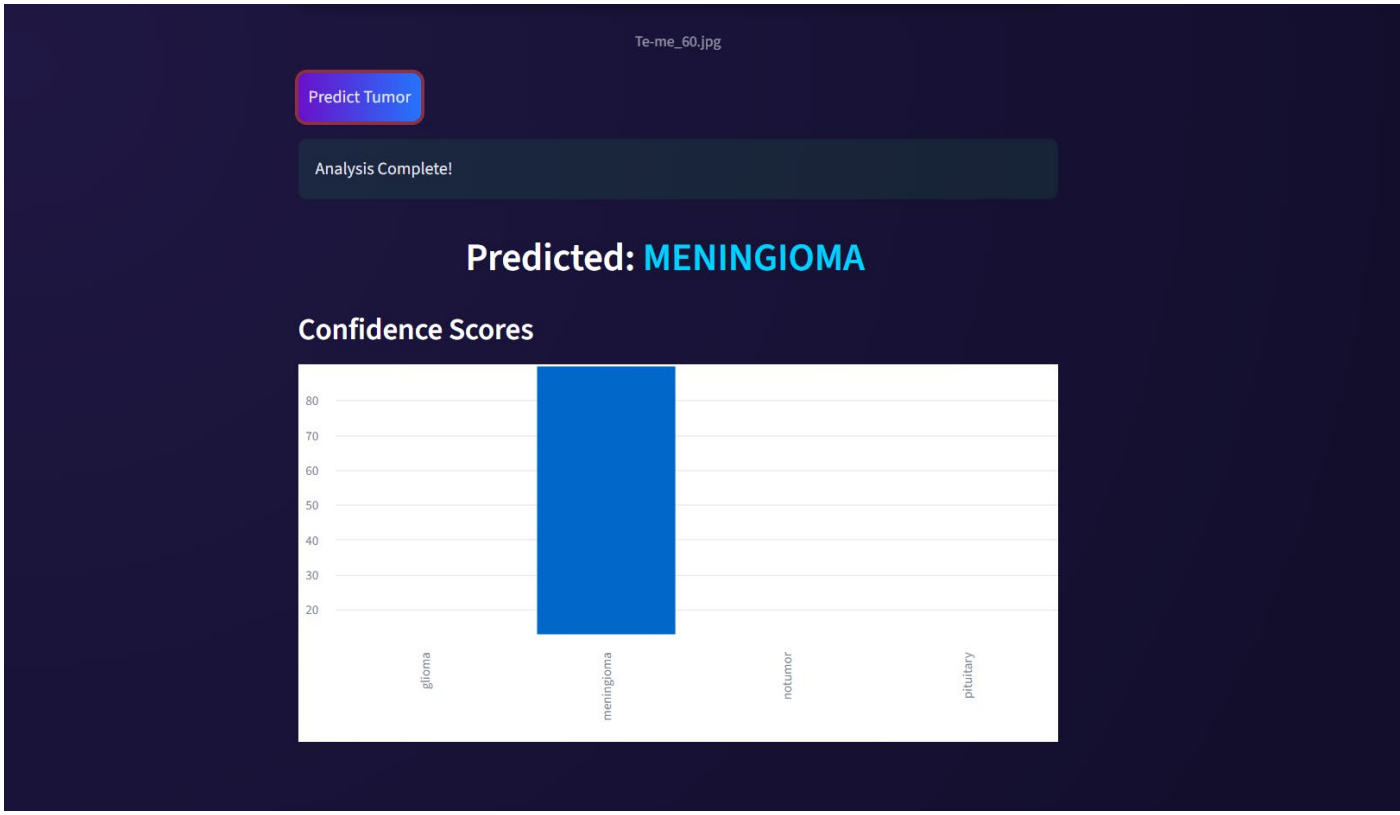

localhost:8501
WhatsApp
YouTube
Erp
MAD
edFlyFees DYP
Alumni MIT
Classroom
Internet Access Login
ChatGPT
Knowledge Pro | Lo...
Login GrayQuest
Grok
Brain Tumor Detection
Upload an MRI scan and let AI analyze it instantly.
Drag & drop your MRI image here or browse files
Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG
Te-me_92.jpg
The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.

To exit full screen, press and hold Esc
Te-me_92.jpg

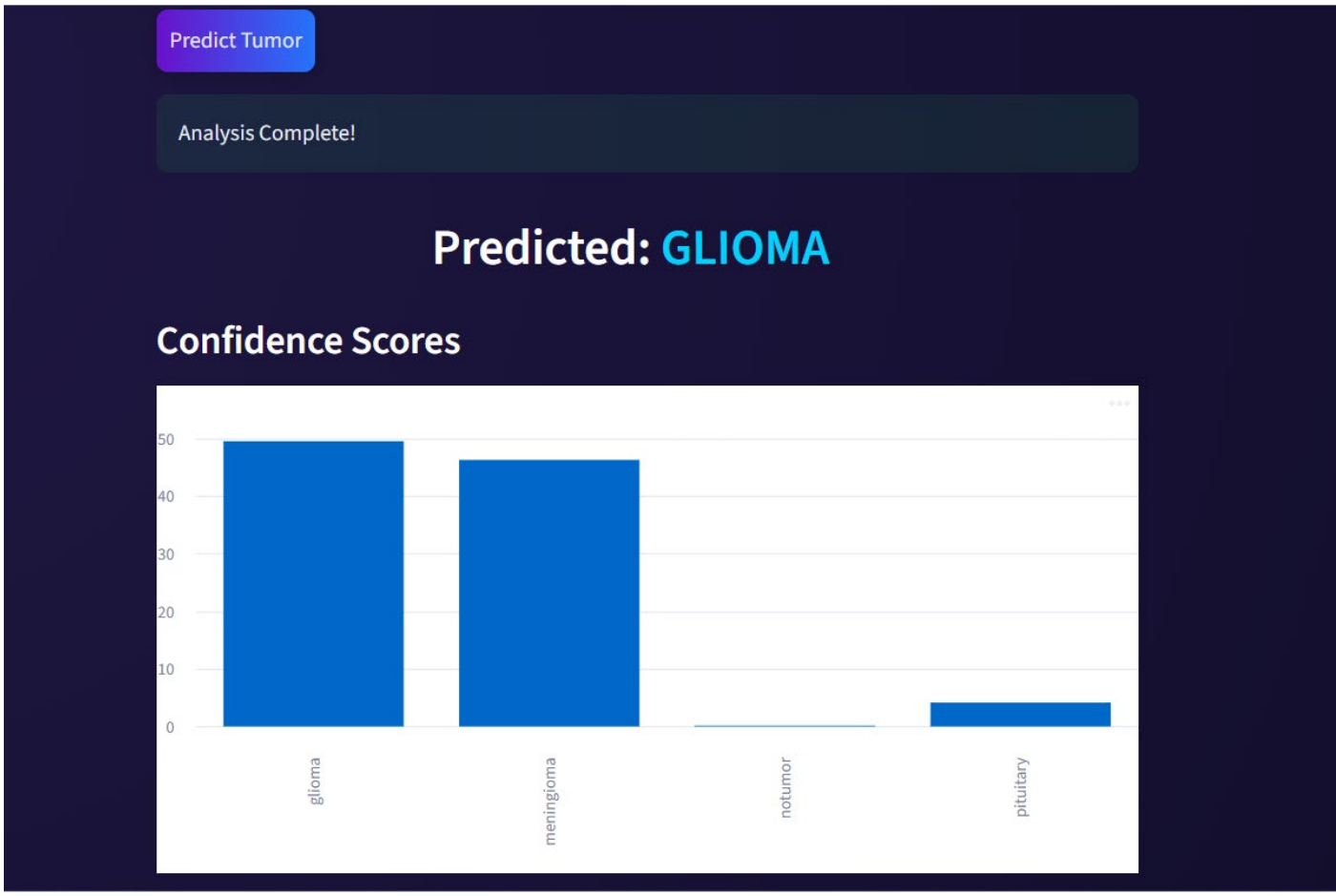


2. Meningioma



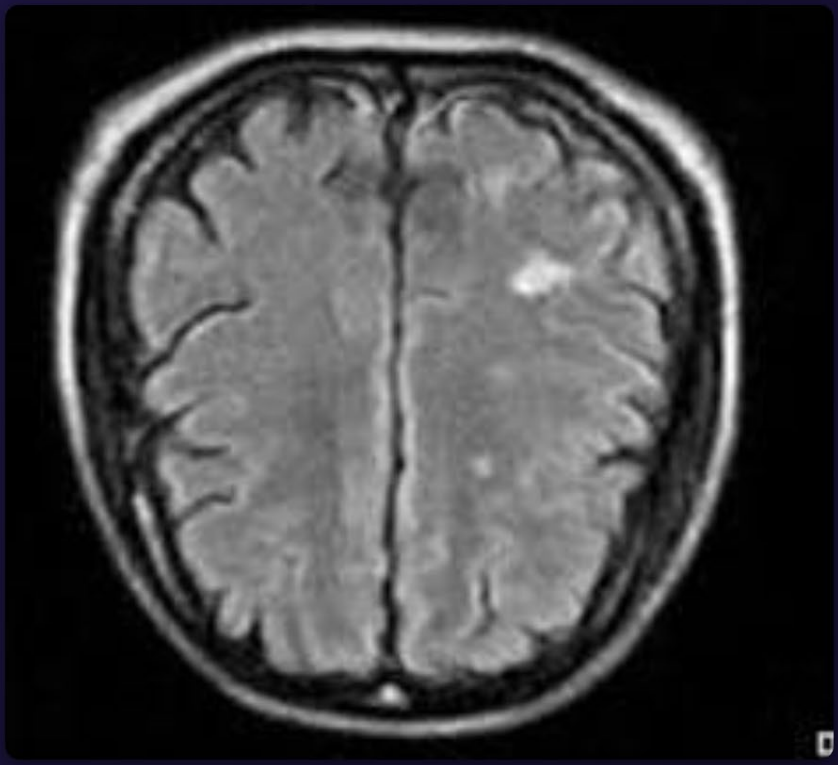


3. Glioma



4. Notumor

The `use_column_width` parameter has been deprecated and will be removed in a future release. Please utilize the `width` parameter instead.

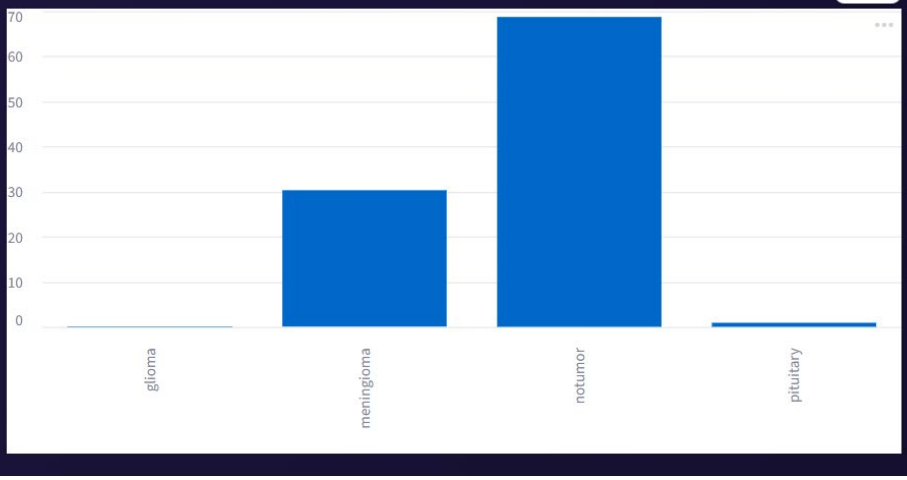


Te-no_98.jpg

Te-no_98.jpg

Predict Tumor

Analysis Complete!

Predicted: NOTUMOR

Confidence Scores